

SHEAF Gate 0 after the Agrimate sitting

what the two-week market does, and what we might change

Working discussion (not a journal submission)

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Abstract

This is a discussion note, not a results paper. Gate 0 is the two-week grain market on which the rest of SHEAF rests: a single crop at a time, historically observed export restrictions imposed exogenously, no substitution among wheat, rice, and maize, and no endogenous policy choice. Coauthors asked how the world price is determined, which object “baseline” refers to, why available grain is withheld rather than sold, and whether the price update should derive from an optimisation problem. We state the two-week recursion in full — in words, then as equations with every symbol defined at the point of use and carrying explicit units — and set each proposed alternative against the equation it would replace and the measurement that would distinguish it. Four figures. Nothing is decided here. Cross-commodity substitution and the restriction game remain deferred until this market is one we would publish as stated.

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1 Purpose and scope

SHEAF is intended as a model in which governments choose export restrictions endogenously and consumers substitute across grains. Neither mechanism is the subject of this note. Before they are added, each crop’s market must be defensible in isolation: prices must move in the correct months, stocks must be compared at the month USDA reports them, and historically observed restrictions must reduce export offers in the countries that imposed them. That prior claim is Gate 0.

Snapshot scores for 2006–11 exist. They are not treated as settled. Following the sitting with the Agrimate authors (Kuhla et al., 2025), the question is whether the *objects underlying* those scores are ones we are prepared to publish. If they are not, adding cross-commodity substitution and an endogenous restriction game on top leaves the foundation weak even where the correlations are acceptable.

The crisis host runs on a two-week step, 24 per year, matching Agrimate’s clock. An annual spatial-equilibrium formulation remains in the repository but is parked; it is a different host, and it produced the wrong sign for the 2007/08 wheat episode.

Structure. Section 2 defines every symbol with its unit and parameter class. Section 3 is the substance: the two-week recursion in words, then as equations, with each symbol restated where it appears so that the equations can be read without turning back. Section 4 takes the sitting’s questions one at a time, stating what the model does now and what each alternative would require. Section 5 gives the measurements that would settle them, and §6 proposes an order of discussion. Four figures are included; alternatives that modify a single equation are described in that equation rather than redrawn.

The standard being applied. Every term used at the sitting should name a specific object in this model, and what we write in a paper should match what the code executes. Known residuals — wheat 2007/08 still high, world consumption not tracking USDA, many exporters resting on the safety-stock floor — remain open until a change is directed at the mechanism that generates them. The 2007/08 window is not a fitting target.

2 Symbols, units, and parameter classes

Countries are indexed by i (importer j where a bilateral pair is needed) and two-week periods by t , with $\Delta t \approx 15.2$ days and exactly two periods per calendar month. Quantities are in million tonnes (MMT); flows are MMT per period. Prices are in constant 2010 US dollars per tonne. Shares are dimensionless and sum to one over the relevant index.

One notational collision is worth flagging: $S_{i,t}$ denotes a *stock* in MMT, while S_{ij} denotes the FAOSTAT *source share* of importer j from exporter i , a dimensionless fraction. The trade

matrices supply bilateral structure only; all traded volumes are determined by the offer and import-demand equations.

Every symbol below is also defined in place at the equation where it first appears, so §3 can be read without returning here.

2.1 State, forcing, and derived quantities

Table 1: Quantities. Stocks and harvests in MMT; prices in constant 2010 \$/t.

Symbol	Definition	Unit
<i>State</i>		
p_t	World price of the crop. Demand in period t uses p_{t-1} .	\$/t
$q_{i,t}$	Offer (export) price posted by exporter i .	\$/t
$S_{i,t}$	Carry-in stock held by country i at the start of period t .	MMT
<i>Forcing</i>		
$H_{i,t}$	Realised harvest arriving in period t .	MMT / step
$H_{i,t}^{\text{seas}}$	Climatological harvest: window-mean production distributed over the calendar.	MMT / step
$\tau_{i,t}$	Export restriction intensity from the AMIS record; the fraction of the offer removed (ban ≈ 0.95 , export tax ≈ 0.50).	fraction
$C_{i,t}^{\text{flex}}$	Food and feed demand evaluated at the reference price p_0 .	MMT / step
$C_{i,t}^{\text{ind}}$	Price-inelastic industrial demand (US maize ethanol residual; zero elsewhere).	MMT / step
C_i^{ann}	Annual domestic use, used to scale stock targets and capacity.	MMT / year
p_0	Reference price: start-year mean of the World Bank Pink Sheet, deflated to 2010 dollars.	\$/t
<i>Derived within the period</i>		
$\text{avail}_{i,t}$	Grain available, $S_{i,t} + H_{i,t}$.	MMT
$d_{i,t}$	Desired demand of country i , Eq. (1).	MMT / step
$H_{i,t}^{\text{exp}}$	Expected harvest, Eq. (2).	MMT / step
$T_{i,t}$	Target stock: lean-season cover plus safety stock, Eq. (3).	MMT
$L_{i,t}$	Lean-season cover: expected consumption shortfall to the next world harvest pulse.	MMT
s_i	Safety stock, $\sigma^{\text{stu}} C_i^{\text{ann}}$.	MMT
$O_{i,t}$	Export supply offered after restriction, Eq. (4).	MMT / step
$D_{i,t}$	Import demand: consumption deficit plus partial rebuilding, Eq. (5).	MMT / step
A_{ij}	FAOSTAT destination share of exporter i to importer j ; $\sum_j A_{ij} = 1$.	fraction

Symbol	Definition	Unit
S_{ij}	FAOSTAT source share of importer j from exporter i ; $\sum_i S_{ij} = 1$.	fraction
$\tilde{S}_{ij,t}$	Importer source shares, CES-reweighted toward low offer prices, Eq. (6).	fraction
$X_{ij,t}$	Bilateral shipment from i to j , Eq. (7).	MMT / step
$X_{i,t}^{\text{out}}, X_{i,t}^{\text{in}}$	Total exports and imports of country i .	MMT / step
$C_{i,t}^{\text{used}}$	Realised consumption after trade, Eq. (8).	MMT / step
$W_{i,t}$	Storage capacity, Eq. (10). Excess above it is physically drawn down, not charged a carrying cost.	MMT
$f_{i,t}$	Fill rate: share of exporter i 's offer that shipped.	fraction
b_t	Blockage: share of world import demand whose preferred suppliers are restricted.	fraction
$u_t, \Delta u_t$	Share of import demand unmet, and its excess over the twin.	fraction
$\mathcal{F}_t$	Market-accessible stock, Eq. (12).	MMT
$\mathcal{F}_t^{\text{twin}}$	The same measure in the unshocked twin run.	MMT
r_t	Scarcity ratio $\mathcal{F}_t^{\text{twin}}/\mathcal{F}_t$, regularised, Eq. (13).	1
p_t^{tr}	Shipment-weighted mean offer price.	\$/t
p_t^{scar}	Scarcity price, Eq. (14).	\$/t
p_t^*	Within-period price target, Eq. (15).	\$/t

2.2 Parameters and current values

Parameters are classified as **L** (literature-sourced: elasticities, stock-to-use norms), **S** (definitional — part of the model's specification, such as the clock, the restriction mapping, and the calendars), or **R** (reduced-form: the sign is economically motivated, the magnitude is not a structural elasticity, and the value is not retuned crisis by crisis).

Parameters carrying the unit “per step” are gains in the discrete two-week update. They are not continuous-time rates and should not be divided by Δt to obtain a physical frequency.

Table 2: Current default values. “Per step” refers to the two-week update.

Symbol	Wheat	Maize	Rice	Class	Unit
ε own-price elasticity	−0.15	−0.25	−0.20	L	elasticity
σ^{stu} safety stock-to-use	0.20	0.16	0.18	L	years of use
σ^{max} capacity stock-to-use	0.28	0.18	0.22	L	years of use
n^{pipe} working inventory	12	12	0	S	steps
λ stock-adjustment rate	0.08	0.08	0.08	R	per step
λ_W capacity drawdown	0.08	0.08	0.08	R	per step
η scarcity elasticity	1.00	0.85	0.95	R	inverse elast.
ρ price smoothing	0.65	0.65	0.65	R	weight
ω weight on offer prices	0.70	0.80	0.72	R	weight
κ_u unmet-demand weight	2.5	2.5	2.5	R	1

Symbol	Wheat	Maize	Rice	Class	Unit
κ_b blockage weight	4.0	4.0	4.5	R	1
α offer-price gain	0.15	0.15	0.15	R	per step
θ neutral fill rate	0.70	0.70	0.70	R	fraction
γ origin-substitution	1.25	1.25	1.25	R	1
β pull toward p_t	0.18	0.18	0.18	R	weight
α_r rival markup	0.80	0.80	0.80	R	per step
ϕ weight on current harvest	0.55	0.50	0.55	R	weight
ν residual pool	0.15	0.15	0.15	S	fraction
world harvest pulse	0.12	0.12	0.12	S	fraction of year

Offer prices are bounded to $[0.45p_0, 2.8p_0]$ and the world price to $[60, 1200]$ \$/t. The rival markup α_r was set by the requirement that maize export restrictions, applied in isolation, must not *reduce* the world price — a sign condition, not a fit to the 2007/08 peak.

3 The two-week market

Each step advances the state $(p_{t-1}, q_{i,t-1}, S_{i,t})$ — last period’s world price, each exporter’s offer price, and carry-in stocks — given that period’s harvest, policy, and demand shifters. The order of operations is fixed and there is no optimisation nested inside the step:

1. **Demand.** Each country forms desired domestic use. Food and feed demand is price-responsive with a short-run elasticity; the industrial (ethanol) component is not. Demand responds to the *previous* period’s world price, and there is no separate domestic price, so an export ban does not lower consumer prices in the restricting country.
2. **Precautionary cover.** Each country targets stocks sufficient to cover consumption until the next harvest, plus a safety margin. This is a target-stock rule, not an intertemporal arbitrage condition.
3. **Export supply and import demand.** Surplus above desired use and the cover target is offered for export and then reduced by the prevailing AMIS restriction. Importers post a food-deficit requirement plus partial rebuilding toward their own cover target.
4. **Bilateral allocation.** Trade follows the observed FAOSTAT partner structure, reweighted toward cheaper offer prices, with a residual pool when preferred links cannot clear.
5. **Stock accounting.** Consumption and shipments are settled; stocks above physical capacity are drawn down.
6. **Offer-price adjustment.** Exporters raise offer prices when their supply clears, cut them when it does not, and mark up when importers’ preferred suppliers are restricted.
7. **World price.** The world price moves toward a target that combines the trade-weighted offer price with a scarcity term measured against an unshocked counterfactual, and is then smoothed.

Three comparisons are worth stating at the outset, because the sitting raised each. Agrimate solves per-region constrained programmes at every step (Kuhla et al., 2025); Gate 0 does not. Competitive storage in the Wright–Williams tradition holds grain when the expected price increase exceeds the cost of carry (Wright and Williams, 1982; Williams and Wright, 1991); Gate 0’s cover rule contains no such arbitrage condition. Spatial price equilibrium determines price as the multiplier on regional balance (Samuelson, 1952; Takayama and Judge, 1971); Gate 0 determines it by an explicit adjustment rule.

The *twin* is a parallel run of the identical mechanism under climatological harvest, mean food and feed demand, no industrial excess, and no restrictions. Its price path is not scored. It supplies the counterfactual against which scarcity is measured, so “tight” means tight relative to a normal year rather than relative to an arbitrary constant.

3.1 Demand and precautionary cover

Desired domestic use is isoelastic in the lagged world price for the flexible (food and feed) component and inelastic for the industrial component:

$$d_{i,t} = C_{i,t}^{\text{flex}} \left(\frac{p_{t-1}}{p_0} \right)^\varepsilon + C_{i,t}^{\text{ind}}, \quad (1)$$

where $d_{i,t}$ is desired demand of country i in period t (MMT per step), $C_{i,t}^{\text{flex}}$ is food-and-feed demand evaluated at the reference price (MMT per step), $C_{i,t}^{\text{ind}}$ is price-inelastic industrial use (MMT per step; the US maize ethanol residual, zero elsewhere), p_{t-1} is the previous period’s world price (2010 \$/t), p_0 is the reference price (2010 \$/t), and $\varepsilon < 0$ is the short-run own-price elasticity of demand.

Expected harvest blends the current realisation with climatology:

$$H_{i,t}^{\text{exp}} = \phi H_{i,t} + (1 - \phi) H_{i,t}^{\text{seas}}, \quad (2)$$

where $H_{i,t}^{\text{exp}}$ is expected harvest (MMT per step), $H_{i,t}$ is realised harvest (MMT per step), $H_{i,t}^{\text{seas}}$ is the climatological harvest path (MMT per step), and $\phi \in [0, 1]$ is the weight on current information. Expectations are adaptive in this single sense; there is no multi-year forecast.

The stock target is lean-season cover plus a safety margin:

$$T_{i,t} = L_{i,t} + s_i, \quad s_i = \sigma^{\text{stu}} C_i^{\text{ann}}, \quad (3)$$

where $T_{i,t}$ is target stock (MMT), $L_{i,t}$ is the shortfall of expected harvest against consumption over the interval to the next world harvest pulse (MMT), s_i is the safety stock (MMT), σ^{stu} is the target stock-to-use ratio (years of use), and C_i^{ann} is annual domestic use (MMT per year). The horizon is defined by the arrival of a fixed fraction of the annual *world* harvest, so it is common across countries.

3.2 Export supply, import demand, and allocation

Export supply is residual surplus, net of policy:

$$O_{i,t} = \max(0, \text{avail}_{i,t} - d_{i,t} - T_{i,t}) (1 - \tau_{i,t}), \quad (4)$$

where $O_{i,t}$ is export supply offered (MMT per step), $\text{avail}_{i,t} = S_{i,t} + H_{i,t}$ is grain available (MMT), $S_{i,t}$ is carry-in stock (MMT), and $\tau_{i,t} \in [0, 1]$ is the export restriction intensity from the AMIS policy record (a ban maps to ≈ 0.95 , an export tax to ≈ 0.50).

Import demand combines an immediate deficit with partial rebuilding:

$$D_{i,t} = \underbrace{\max(0, d_{i,t} - \text{avail}_{i,t})}_{\text{consumption deficit}} + \lambda \max(0, T_{i,t} - \max(0, \text{avail}_{i,t} - d_{i,t})), \quad (5)$$

where $D_{i,t}$ is import demand (MMT per step) and λ is the partial-adjustment rate toward the stock target (per step). Equation (5) is a stock-adjustment specification, not a solution to a dynamic programme.

Bilateral allocation is Armington on a prescribed network ([Armington, 1969](#)). Importer source shares are the CES expenditure-minimising mix over origins,

$$\tilde{S}_{ij,t} \propto S_{ij} \left(\frac{p_0}{q_{i,t-1}} \right)^\gamma, \quad \sum_i \tilde{S}_{ij,t} = 1, \quad (6)$$

where S_{ij} is the observed FAOSTAT source share of importer j from exporter i (dimensionless; note S_{ij} is a share, whereas $S_{i,t}$ is a stock), $q_{i,t-1}$ is exporter i 's lagged offer price (2010 \$/t), and $\gamma > 0$ is the elasticity of substitution across origins. An earlier draft reweighted destination shares A_{ij} by the same factor and renormalised each exporter's row. That map is the identity: a row-constant cancels, so cheaper origins never gained share. Equation (6) is the importer-side object that actually responds to relative offer prices, and is the first-order condition of a CES expenditure-minimisation problem. Shipments on each link are the short side,

$$X_{ij,t} = \min(O_{i,t} A_{ij}, \tilde{S}_{ij,t} D_{j,t}), \quad (7)$$

where $X_{ij,t}$ is the shipment from i to j (MMT per step) and A_{ij} is the observed destination share of exporter i to importer j (dimensionless). Unmatched supply then meets a fraction ν of unmatched demand through a residual pool. Quantities therefore come from (4)–(5); the trade matrix supplies only the pattern of partners.

3.3 Stock accounting and capacity

Realised consumption is bounded by post-trade availability:

$$C_{i,t}^{\text{used}} = \min(d_{i,t}, \max(0, \text{avail}_{i,t} - X_{i,t}^{\text{out}} + X_{i,t}^{\text{in}})), \quad (8)$$

where $C_{i,t}^{\text{used}}$ is realised consumption (MMT per step), $X_{i,t}^{\text{out}} = \sum_j X_{ij,t}$ is total exports and $X_{i,t}^{\text{in}} = \sum_j X_{ji,t}$ total imports (MMT per step). Closing stock is the residual, less any drawdown of grain held above physical capacity:

$$S_{i,t+1} = S_{i,t+1}^- - \lambda_W \max(0, S_{i,t+1}^- - W_{i,t}), \quad S_{i,t+1}^- = \max(0, \text{avail}_{i,t} - X_{i,t}^{\text{out}} + X_{i,t}^{\text{in}} - C_{i,t}^{\text{used}}), \quad (9)$$

where $S_{i,t+1}$ is closing stock (MMT), $S_{i,t+1}^-$ is closing stock before the capacity adjustment (MMT), $W_{i,t}$ is storage capacity (MMT), and λ_W is the rate at which excess above capacity is drawn down (per step). This term removes grain physically; it is not a pecuniary cost of carry,

and no interest rate appears anywhere in (9). Capacity is

$$W_{i,t} = \sigma^{\max} C_i^{\text{ann}} + \frac{n^{\text{pipe}}}{24} C_i^{\text{ann}} + \mathbf{1}\{n^{\text{pipe}} > 0\} H_{i,t}, \quad (10)$$

where $\sigma^{\max}$ is the maximum stock-to-use ratio (years of use) and n^{pipe} is working inventory in steps (12 for wheat and maize, i.e. six months; 0 for rice, which harvests year-round).

3.4 Offer prices

Exporters adjust offer prices toward market conditions:

$$q_{i,t} = (1 - \beta) q_{i,t-1} \exp\left(\alpha(f_{i,t} - \theta) + \alpha_r b_t \mathbf{1}\{O_{i,t} > 0\}\right) + \beta p_{t-1}, \quad (11)$$

where $q_{i,t}$ is exporter i 's offer price (2010 \$/t), $f_{i,t} = X_{i,t}^{\text{out}}/O_{i,t}$ is the fill rate (dimensionless), θ is the fill rate at which the price is left unchanged, $\alpha > 0$ is the adjustment gain (per step), b_t is the share of world import demand whose preferred suppliers are restricted (dimensionless), $\alpha_r \geq 0$ is the markup unrestricted exporters apply when rivals are blocked, and $\beta \in [0, 1]$ is the weight pulling offer prices toward the world price. Offer prices are bounded to $[0.45p_0, 2.8p_0]$.

Equation (11) is an adjustment rule: excess demand raises the price, excess supply lowers it. It is not the first-order condition of an exporter's profit-maximisation problem, and α, θ, β are not structural markups.

3.5 The world price

Define market-accessible stocks — total stocks net of lean-season requirements and of grain sequestered behind restrictions:

$$\mathcal{F}_t = \sum_i S_{i,t+1} - \sum_i L_{i,t} - \sum_i \tau_{i,t} \max(0, S_{i,t+1} - T_{i,t}), \quad (12)$$

where $\mathcal{F}_t$ is market-accessible stock (MMT). The third term matters: without it, a restriction would register as abundance. Let $\mathcal{F}_t^{\text{twin}}$ be the same quantity in the twin, and define the scarcity ratio

$$r_t = \frac{\mathcal{F}_t^{\text{twin}} + f_t}{\mathcal{F}_t + f_t}, \quad f_t = 0.05 \sum_i s_i + \max(0, -\min(\mathcal{F}_t, \mathcal{F}_t^{\text{twin}})), \quad (13)$$

where f_t is a regulariser (MMT) keeping (13) finite and positive. It is worth being explicit that f_t is neither small nor constant, because it changes how (13) should be read. Its first term is 5% of world safety stock: 6.4 MMT for wheat, 6.5 for maize, 3.9 for rice, which is 4.8%, 3.1% and 10.3% of mean accessible stock in the scored window. Against the unregularised ratio $\mathcal{F}_t^{\text{twin}}/\mathcal{F}_t$ it biases r_t by 2.2% on average for wheat, 3.4% for maize and 35.4% for rice. The second term exists because $\mathcal{F}_t$ as defined in (12) can be negative — the world can hold less than its own lean-season requirement — which happens at 1 of 144 steps for maize and 58 of 144 for rice. Where that occurs the unregularised ratio is meaningless, so for rice in particular r_t is materially

a regularised quantity rather than a ratio of stocks. The price target and its updating are

$$p_t^{\text{scar}} = p_0 r_t^\eta (1 + \kappa_u \Delta u_t + \kappa_b b_t), \quad (14)$$

$$p_t^* = \omega p_t^{\text{tr}} + (1 - \omega) p_t^{\text{scar}}, \quad (15)$$

$$p_t = \rho p_{t-1} + (1 - \rho) p_t^*, \quad (16)$$

where p_t^{scar} is the scarcity price (2010 \$/t), $\eta > 0$ is the elasticity of price with respect to the inverse of accessible stocks, $\kappa_u, \kappa_b \geq 0$ weight unmet demand and restriction blockage, p_t^{tr} is the shipment-weighted mean offer price (2010 \$/t), $\omega \in [0, 1]$ is the weight on realised trade prices, p_t^* is the within-period price target (2010 \$/t), and $\rho \in [0, 1]$ is the smoothing parameter. The world price is bounded to $[60, 1200]$ \$/t.

Two details of (14)–(15) are easy to state imprecisely and are worth pinning down. Before them, one omission. The trade-weighted offer price is undefined when nothing ships, and the implementation resolves that by falling back to the previous world price,

$$p_t^{\text{tr}} = \begin{cases} \frac{\sum_i q_{i,t} x_{i,t}}{\sum_i x_{i,t}}, & \sum_i x_{i,t} > \epsilon, \\ p_{t-1}, & \text{otherwise,} \end{cases} \quad (17)$$

where $x_{i,t}$ is the quantity country i actually ships in period t (MMT) and $\epsilon = 10^{-12}$ MMT. This is not a removable convention. As world shipments vanish the weighted mean tends to the offer price of whichever exporter still ships, which need not be p_{t-1} , so (17) is genuinely discontinuous at the threshold: perturbing the price across it moves p_t^{tr} by up to 12 \$/t. The threshold is not crossed on the scored paths, but it is reachable under price perturbation at a handful of steps, all of them where accessible stock is small or negative.

Now the two details. First, the unmet-demand term is one-sided,

$$\Delta u_t = \max(0, u_t - u_t^{\text{twin}}), \quad (18)$$

where u_t is the share of world import demand left unmet and u_t^{twin} its value in the twin. Unmet demand below the twin therefore cannot lower the price, whereas unmet demand above it raises the price. The truncation binds: $u_t < u_t^{\text{twin}}$ at 92 of 144 steps for wheat, 59 for maize and 100 for rice, so this is the usual case rather than an edge case. Second, the offer prices in (17) are this period's $q_{i,t}$ from (11), not the lagged $q_{i,t-1}$ that entered the allocation in (6). Offer prices are therefore updated twice within one period's causal chain: once to allocate trade, once to price it.

Equation (14) is inverse-demand in form: price rises as accessible stocks fall relative to normal, with elasticity η . The exponent is symmetric, so a surplus lowers the price and food and feed demand absorbs it; muting the abundance side left gluts sitting in store. Equations (15)–(16) then mix that term with realised trade prices and smooth the result, so the world price is not required to lie on the inverse-demand schedule within the period. That is the honest gap, and it is the subject of §4.

3.6 Properties, and one that is enforced rather than derived

Reference identity. Suppose the crisis run matches the twin in accessible stocks, unmet import demand, and blockage: $\mathcal{F}_t = \mathcal{F}_t^{\text{twin}}$, $\Delta u_t = 0$, $b_t = 0$. Then $r_t = 1$ and (14) gives

$p_t^{\text{scar}} = p_0$. The implementation additionally sets $p_t^* = p_0$ outright in this case, by an explicit conditional on those three quantities. That conditional is doing real work, and an earlier draft of this note was wrong to describe the identity as following from the algebra.

The algebra alone gives $p_t^* = \omega p_t^{\text{tr}} + (1 - \omega)p_0$, which equals p_0 only if $p_t^{\text{tr}} = p_0$, i.e. only if offer prices rest at the reference. They do not. In a matched run — climatological harvest, mean food and feed demand, no industrial excess, $\tau \equiv 0$ — realised fill sits well below the target $\theta = 0.70$, averaging 0.54 for wheat, 0.32 for maize and 0.62 for rice, so (11) pushes offer prices down every period even with nothing happening. The trade-weighted offer price settles near $0.81p_0$ for wheat and $0.71p_0$ for maize. With the conditional removed, the world price in a matched run drifts 25% for wheat, 34% for maize and 19% for rice, against the 2% tolerance at which the twin-identity test is written. Re-targeting θ to the realised matched-run fill does not repair this: the drift falls only to about 19% for wheat and 20% for maize, because the stationarity condition is per-exporter and the realised fill is highly dispersed across countries and periods.

So the honest statement is that (11) has no rest point at p_0 in a market with nothing happening, and the conditional is what makes the reference exact. Two things bound how much this matters. It changes no scored result: the conditional fires at 0 of 144 steps in the official harvest-plus-restrictions run and 0 of 144 in the harvest-only run, because any harvest anomaly breaks the equality immediately. But it is load-bearing for the model’s own robustness tests, firing at all 144 steps of the matched run that the twin-identity test uses and at 24–32 steps of the restriction-only run that the restriction-raises-price test uses as its baseline. Whether to keep it and document it, or to rebuild (11) so that a quiet market is a genuine fixed point, is a live design question and is taken up in §4.

Note finally that $\mathcal{F}_t^{\text{twin}}$ oscillates seasonally, so the reference is a trajectory rather than a constant.

Twin closure. During the twin run the price is held at p_0 , so $\mathcal{F}_t^{\text{twin}}$ reflects physical availability and does not feed back through (1). The twin is a control, not a second priced market.

Mass balance. Availability is exhausted by consumption, net trade, the change in stocks, and the capacity drawdown in (9). The drawdown is part of the specification and is reported, not absorbed silently.

4 Questions raised at the sitting

Each subsection states a question raised by a coauthor, what Gate 0 currently does, what the alternative would require, and the measurement that would distinguish them. No option is selected here.

It is worth separating two classes of alternative. Some are reparameterisations or single-equation substitutions within the two-week recursion of §3. Others replace the host: per-agent constrained optimisation as in Agrimate (Kuhla et al., 2025), or annual spatial price equilibrium (Samuelson, 1952; Takayama and Judge, 1971). The second class is not an edit. Figure 4 shows that division.

4.1 Which object is “the baseline”?

The term was used for four distinct objects, and the ambiguity is worth resolving before anything else.

The *reference price* p_0 is the start-year mean of the World Bank Pink Sheet in constant 2010 dollars, where p_0 enters (1) as the price at which $C_{i,t}^{\text{flex}}$ is defined and (14) as the level the scarcity price returns to. The *climatological harvest* $H_{i,t}^{\text{seas}}$ is window-mean production distributed over the calendar, and enters expectations through (2). The *spin-up* initialises stocks $S_{i,t}$ from 2005 USDA levels and runs the unshocked recursion for two years, so that opening stocks are dynamically consistent rather than a raw data snapshot. The *twin* is a full parallel run of the same recursion under H^{seas} , mean food and feed demand, no industrial excess, and $\tau_{i,t} = 0$, which supplies $\mathcal{F}_t^{\text{twin}}$ in (13). Scarcity is measured against the twin, not against p_0 alone.

Agrimate spins up for roughly four years. Lengthening ours is a sensitivity test on initial conditions, not a change of theory. Replacing the twin with an annual market-clearing reference is a change of host (Fig. 4D).

4.2 Why is available surplus not fully offered each period?

Because (4) withholds the target stock $T_{i,t}$ defined in (3). A country consumes $d_{i,t}$, retains lean-season cover plus safety stock, and offers the residual. No interest rate and no expected-price comparison appear: there is no condition of the form $\mathbb{E}_t[p_{t+1}] \geq (1+r)p_t + c$ that a competitive storer would impose, where r is the interest rate and c the physical cost of carry (Wright and Williams, 1982; Williams and Wright, 1991; Deaton and Laroque, 1992). Slide 48 of the deck is the offer-price update (11), not a cost of carry; that reading should be confirmed with the authors.

Agrimate’s commercial supplier does solve a store-versus-sell problem from expected profit, conditional on announced restrictions (Kuhla et al., 2025). Adopting that is Fig. 4C, not a modification of (3). Note also that Agrimate separates commercial storage from a strategic-reserve agent following a partial-adjustment refill rule; SHEAF’s Gate 2 food-security trigger is not that second agent, and the two should not be equated when comparing the storage layers. The intermediate option is an explicit arbitrage margin: hold above $T_{i,t}$ only when $\mathbb{E}_t[p_{t+1}] - p_t$ exceeds a per-step carry cost. This introduces both a price expectation and a carry parameter, neither of which the model currently has. A related experiment — country-specific safety stock ratios σ_i^{stu} — improved US wheat stocks but inflated Chinese maize stocks by roughly a factor of three, and was not retained.

4.3 How is the world price determined?

By (14)–(16): a shipment-weighted offer price combined with a scarcity term and then smoothed. The system is a discrete map, not a differential equation; the only relaxation-like element is the offer-price adjustment (11).

Three distinct requests were compressed into “there should be an optimisation.” The first is *agent optimisation*: per-region programmes at every step, i.e. Agrimate (Fig. 4C). The second is *market clearing*: price as the multiplier on a balance constraint in an annual spatial equilibrium (Fig. 4D) — the host that produced the wrong sign for 2007/08. The third is the observation that Gate 0 may already contain an optimisation principle, since (14) has the form of inverse demand over accessible stocks with elasticity η .

Three gaps remain between that form and a genuine equilibrium, and they should be stated rather than papered over. Demand (1) uses p_{t-1} , so quantities entering trade are not consistent with the price the same period produces. Blending and smoothing in (15)–(16) means p_t does not lie on the inverse-demand schedule within the period. And (3) is a target-stock rule, so storage is not a first-order condition. Writing these three sentences accurately is cheaper, and more defensible, than attaching an $\arg \max$ to the existing recursion. TWIST likewise did not solve a storage Euler equation (Schewe et al., 2017).

Two alternatives remain inside the two-week host. The first is contemporaneous consistency: replace p_{t-1} with p_t in (1) and solve the resulting scalar fixed point $p_t = \mathcal{G}(p_t)$ each period, where $\mathcal{G}$ is the composition of (1)–(16). That is one univariate root-find per crop per step, not a system of programmes. The second is a structural offer rule: replace the exponential adjustment in (11) with a markup over marginal cost, retaining the inverse-demand-plus-offers structure of (15).

4.4 Should a quiet market be a fixed point?

This question was not asked at the sitting. It emerged from checking the equations in §3 line by line against the implementation, and it is the most substantive thing that check produced, so it is recorded here rather than buried.

The design intent is that a market with no harvest anomaly, no restriction and no demand shifter should sit at the reference price. The implementation delivers that intent with an explicit conditional. The underlying law does not: realised fill in a matched run averages 0.54, 0.32 and 0.62 for wheat, maize and rice against a target $\theta = 0.70$, so (11) pushes offer prices down every period in a market where nothing is happening, and the world price drifts 19–34% once the conditional is removed. Three responses are available.

Keep it and document it. Defensible, and cheapest. The conditional fires at 0 of 144 steps in both scored legs, so no published number depends on it. What must change is the claim: the reference is a modelling axiom enforced at one point in the map, not a property derived from the price law. The cost is that a referee who reads (11) and asks what it does in a quiet market gets an answer we would rather not give.

Re-target θ . Tested and rejected. Setting θ to the realised matched-run fill leaves 19–20% drift, because stationarity in (11) is per-exporter while the realised fill is widely dispersed, and it degrades the maize price correlation from +0.71 to +0.65. The mean fill is the wrong object to match.

Rebuild (11) so the quiet market is a genuine rest point. The principled option, and the one that connects to the optimisation question in §4: an offer rule derived from an exporter problem would have its rest point determined by that problem rather than by a free target-fill parameter. This is a real change with score consequences across all three crops, and it should not be undertaken on the strength of this diagnostic alone.

The recommendation is the first for now and the third as the candidate if the offer-price law is revisited for other reasons. What should not happen is quietly keeping the conditional while continuing to describe the identity as algebraic.

4.5 Who trades with whom, and at which price?

Traded quantities are determined by (4) and (5); the FAOSTAT matrices A_{ij} and S_{ij} supply only the bilateral pattern, reweighted by (6). A restriction reduces $O_{i,t}$ directly, whereas realised

shipments $X_{ij,t}$ are the short side of (7); Russian wheat offers in late 2010 therefore collapse by far more than Russian shipments do. The phrase “sold to each market” in the notes most likely refers to Agrimate’s purchaser agent rather than to its storer.

A restriction does not currently generate a domestic price below p_t . Introducing a domestic price $p_{i,t}^{\text{dom}}$ distinct from the world price would matter for a consumer-welfare analysis and would connect to the price-insulation literature (Martin and Anderson, 2012); it is not required to reproduce the single World Bank world price series that Gate 0 is scored against.

4.6 Is demand linear, and what counts as industrial use?

Demand is isoelastic, not linear: (1) is $C^{\text{flex}}(p_{t-1}/p_0)^\varepsilon$, so the elasticity ε is constant and the price response is multiplicative. The linear system $D = a - Mp$ belongs to the parked annual host, where it is the quadratic-programme demand block; it is not the crisis specification. The two should not be conflated when comparing SHEAF against the annual formulation.

The industrial component $C_{i,t}^{\text{ind}}$ is constructed as US maize food-seed-industrial use minus its 2000–04 mean, distributed evenly across periods, and is zero for every other country and crop. It is the post-Renewable-Fuel-Standard ethanol expansion treated as an exogenous, price-inelastic demand shifter (U.S. Congress, 2005; Abbott et al., 2011; Roberts and Schlenker, 2013), not an endogenous biofuel sector.

For official scores, $C_{i,t}^{\text{flex}}$ is the window-mean food and feed level divided by 24 rather than the year-by-year USDA path. That choice matches Agrimate’s treatment and is the reason simulated world consumption does not track the USDA series; using the observed annual path is available as a sensitivity but would change what the official split means.

4.7 Should a surplus lower the price, and on what frequency?

It does, within the same period. The exponent η in (14) is symmetric: when accessible stocks exceed the twin, $r_t < 1$ and the scarcity price falls below p_0 , which raises food and feed demand through (1) in the following period. An earlier specification muted the abundance side, and the consequence was that surpluses accumulated in storage instead of clearing.

The frequency question is separate and worth stating plainly. Abundance in grain markets is resolved over a marketing year, not a fortnight, and Gate 0 has no mechanism that operates at that frequency: there is no acreage response, and ϕ in (2) looks forward only to the next harvest pulse. What the two-week recursion can represent is the within-season absorption of a surplus through consumption and stock accumulation. Multi-year adjustment is outside its scope, and neither TWIST (Schewe et al., 2017) nor Agrimate (Kuhla et al., 2025) makes production endogenous either.

4.8 What is the expectational horizon?

Short, and confined to one place. Equation (2) places weight $\phi \approx 0.5$ on the current harvest realisation and the remainder on climatology, and this expectation enters only the cover target (3). Demand (1) uses the realised lagged price p_{t-1} , which is backward-looking rather than expectational. The lean-season horizon in $L_{i,t}$ is defined by arrival of 12% of the annual *world* harvest and is therefore common across countries, not indexed to national harvest calendars.

The statement “prices stay near the reference unless new information arrives” is the identity established in §3: with $\mathcal{F}_t = \mathcal{F}_t^{\text{twin}}$, $\Delta u_t = 0$ and $b_t = 0$, (14) returns p_0 exactly. One live option

is to index $L_{i,t}$ to national harvest calendars. Rational-expectations storage over a long horizon (Deaton and Laroque, 1992) is the opposite of what was requested and would change the host.

4.9 Is the parameter count defensible?

The concern is legitimate and should be answered by classification and by ablation rather than by assertion. Table 2 labels each parameter as literature-sourced (L), definitional (S), or reduced-form (R). Reduced-form parameters have economically signed effects but are not structural elasticities, and they are not retuned crisis by crisis.

Separately, the model contains several inequality constraints — storage capacity $W_{i,t}$ in (10), the offer-price bounds on (11), the world-price bounds on (16), and the AMIS mapping to $\tau_{i,t}$. Coauthors observed that a sufficiently constrained rule-based system can mimic the outcome of an optimisation. The cheapest empirical response is channel ablation: set $\omega = 1$ or $\omega = 0$ in (15), set $\alpha_r = 0$ or $\kappa_b = 0$ in (11) and (14), and record which channel carries 2007/08 versus 2010/11. That is a decomposition, not a refit.

5 Diagnostics that would distinguish the alternatives

Each item below is a measurement on the current recursion, or on a labelled experiment that does not retune the 2007/08 window. They are stated so that an alternative can be rejected or accepted on evidence rather than on preference.

Properties that already hold, and how. Four are asserted in tests, but they do not all hold for the same reason, and the distinction turned out to matter. When the crisis run matches the twin in accessible stocks, unmet import demand, and blockage, the implementation returns $p_t^* = p_0$ — by the explicit conditional described in §3, not by the algebra of (14)–(15). Any test of this property is therefore a test that the conditional is present and that the matched configuration is genuinely matched; it is not evidence that the price law returns to the reference on its own. A test that would be evidence of the latter is given below. The remaining three are structural: maize export restrictions applied in isolation must not lower the world price, which is the condition that set α_r in (11); availability is exhausted by consumption, net trade, the change in stocks, and the capacity drawdown in (9); and the twin holds price at p_0 , so it exerts no feedback through (1).

Does a quiet market return to the reference? The test worth adding. Disable the conditional, run the matched configuration, and measure the drift of p_t from p_0 . Currently that drift is 25% for wheat, 34% for maize and 19% for rice, so the answer is no: the offer-price law has no rest point at the reference. If (11) is ever rebuilt, this is the measurement that would show it had worked, and it is worth asserting at a stated tolerance so the property cannot silently regress. Note that the existing restriction-raises-price test uses the restriction-only run as its baseline, and the conditional fires at 24–32 of its 144 steps; if the conditional is removed, that baseline drifts and the test’s lift measurement loses its meaning unless it is re-specified.

Contemporaneous demand (option B). Along a scored trajectory, recompute (1) at the price p_t that the period actually produced, and compare it to the $d_{i,t}$ that entered (4). The distribution of $d_{i,t}(p_t) - d_{i,t}(p_{t-1})$ bounds what contemporaneous consistency can do. If it is small,

the fixed point changes labels rather than results; if it is large in 2007/08, the informational structure genuinely differs and the change is substantive.

Offer prices as a markup rule. Regress $\log(q_{i,t}/q_{i,t-1})$ on the fill rate $f_{i,t}$ from (11). The current specification is exponential in $f_{i,t} - \theta$; a markup over marginal cost implies a different relationship, in particular a dependence on cost rather than on excess demand. Establish whether the existing rule is already observationally close to a markup before replacing it.

Storage as intertemporal arbitrage. Under an arbitrage condition, offers should rise with the expected price increase. Under (4) they rise with surplus above $T_{i,t}$. Scatter $O_{i,t}$ against $p_{t+1} - p_t$ as a descriptive baseline, recognising that this is not decisive because cover already covaries with the harvest calendar. The failed experiment with country-specific σ_i^{stu} (Chinese maize stocks inflating threefold) is the relevant prior.

Domestic price insulation. Inspect the state directly: no variable distinct from p_t enters (1) for a restricting country. Any claim about consumer welfare inside a banning country is therefore outside what Gate 0 currently represents (Martin and Anderson, 2012).

National versus world lean-season clocks. Compare the world harvest-pulse horizon in $L_{i,t}$ against each country’s own harvest calendar. Cover held in the month preceding a country’s own harvest is a direct signature of the common world clock.

Channel ablation. Re-run the official harvest-plus-restrictions score with, in turn, $\omega = 1$ (offer prices only), $\omega = 0$ (scarcity only), $\alpha_r = 0$ (no rival markup), and $\kappa_b = 0$ (no blockage term). This attributes 2007/08 and 2010/11 to specific channels without introducing a new specification.

Known residuals under the present rule. These stand regardless of which option is chosen, and should not be used to motivate a change unless the change targets their mechanism. Wheat 2007/08 remains high and restriction-led. World consumption does not track the USDA path, which is expected given that official scores use mean food and feed demand and (1) reduces consumption as price rises. Many exporters sit at the safety-stock floor s_i . Restrictions cut offers substantially more than shipments. Introducing an optimisation under the world price does not automatically resolve any of these, and fitting κ_b to 2007/08 would not constitute standing behind the market.

6 Proposed order of discussion

This is an order for the conversation, not an implementation schedule.

1. **Terminology.** Agree that “baseline,” “offer,” “share,” “ask,” and “world price” each denote one object (Fig. 2, §2). Several apparent disagreements at the sitting were about naming.
2. **Accurate description.** Establish whether (14) already behaves as inverse demand and whether (11) already behaves as a markup. Any text stating that SHEAF contains Agrimate-style optimising agents should be corrected; it does not.

3. **Storage.** Decide between the cover rule (3) (Fig. 3) and an explicit arbitrage condition. Confirm with the Agrimate authors that slide 48 refers to offer prices rather than a cost of carry.
4. **Price formation.** Choose among retaining (16) as it stands, imposing contemporaneous consistency between (1) and p_t , or giving exporters a structural markup. This follows items 2 and 3 so that a markup rule is not combined with a cover rule by accident.
5. **Ablation before respecification.** Run the channel decomposition in §5 before adopting any new official specification.

Three outcomes are possible for each item. If the alternative turns out to be the current mechanism under an accurate label, we retain the equation and correct the description. If it is contemporaneous consistency or a one-equation storage rule, it is a labelled experiment and the corresponding diagnostic in §5 should be run first. If it is per-agent optimisation or annual market clearing, it is a change of host and should be evaluated as such.

Four things are explicitly out of scope: fitting the 2007/08 window, selecting a preferred substitution elasticity, folding the annual formulation back into the two-week recursion, and treating Deaton–Laroque (Deaton and Laroque, 1992) as the standard against which the storage rule is judged, given that neither TWIST (Schewe et al., 2017) nor Agrimate (Kuhla et al., 2025) implements it. Headey’s chronology (Headey, 2011) determines when a later restriction game should act; it does not by itself motivate retuning this market. Questions about how the world price is formed do.

Cross-commodity substitution and the restriction game remain deferred until Gate 0 is a market we would publish as stated.

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Figure 1: Order of operations within one two-week step. Each box is evaluated once; no inner solve.

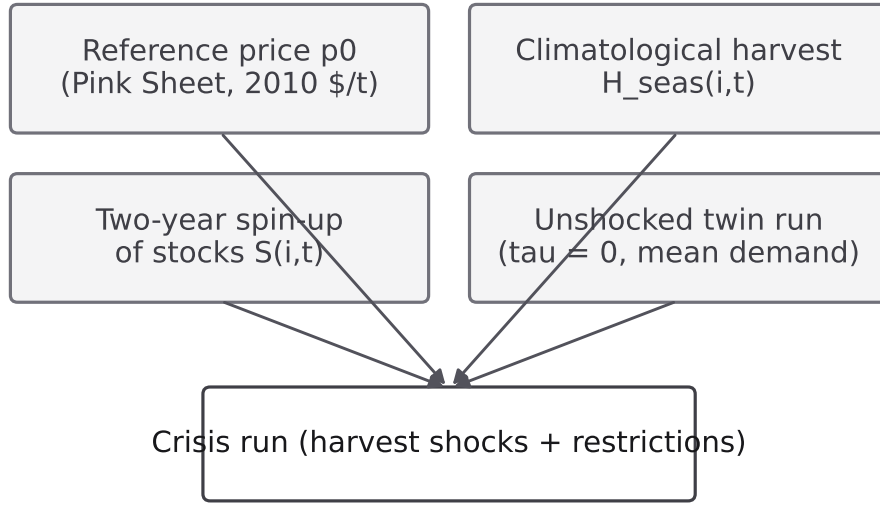


Figure 2: The four objects that have all been called “baseline.” The scarcity term in (14) is measured against the twin.

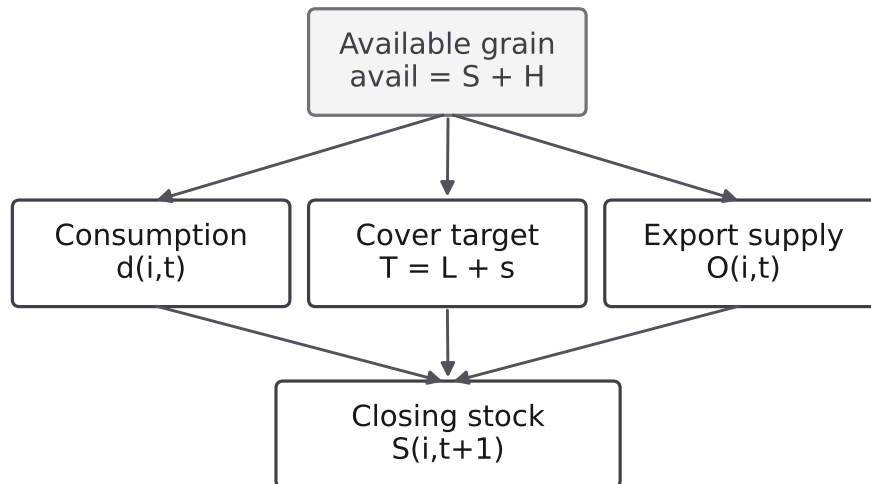


Figure 3: Partition of available grain under (4): consumption, cover, offer. No store-versus-sell programme is solved.

A. Current recursion (Gate 0)

B. Contemporaneous consistency



C. Agent optimisation (Agrimate)

D. Annual spatial equilibrium



Figure 4: Four hosts capable of producing a world price. A is the current recursion. B remains the two-week host with demand and price made contemporaneously consistent. C and D are different models.